SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Network COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

I Suganthan Kumaresan (192424105) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.Suganthan

Annexure A

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

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1. File Transfer (150 MB)

Given:

- File size = 150 MB = **150×10^6 = 150,000,000 bytes**
- Segment size = **1500 bytes**
- Frame overhead = **40 bytes**
- Bandwidth = **50 Mbps**

(a) Number of segments

$$\frac{150{,}000{,}000}{1500} = 100{,}000 \text{ segments}$$

Frames transmitted = 100,000 (1 segment per frame)

(b) Total Data Link overhead

$$100{,}000 \times 40 = 4{,}000{,}000 \text{ bytes} = 4 \text{ MB}$$

(c) Total transmission size

$$150 + 4 = 154 \text{ MB}$$

Bits transmitted:

$$154 \times 8 = 1232 \text{ Mb}$$

Transmission time:

$$\frac{1232}{50} = 24.64 \text{ s}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Overhead = **4 MB**
- Transmission time ≈ **24.64 s**

Reliability

- **Transport Layer:** Segmentation, sequencing, acknowledgements, retransmissions.
- **Data Link Layer:** Framing, error detection (CRC), reliable frame delivery between adjacent nodes.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

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2. Sliding Window

Given

- Window size = 6
- Total frames = 48

Number of windows

$$\frac{48}{6} = 8$$

Retransmissions

1 frame lost per window

$8 \times 1 = 8$

Answer

- Windows = **8**
- Retransmissions = **8**

Flow Control

Flow control prevents the sender from overwhelming the receiver, improving efficiency and reducing packet loss.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

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3. IP Fragmentation

Given

- Packet = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Maximum data per fragment:

$1500 - 20 = 1480$ bytes

Fragments:

$\lceil \frac{12000}{1480} \rceil = 9$

Header overhead:

$9 \times 20 = 180$ bytes

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Network Layer

Uses fragmentation, fragment offsets, identification fields, and reassembly at the destination.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

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4. Sliding Window (Same as Q2)

Answer

- Windows = **8**
- Retransmissions = **8**

Flow Control

Matches sender speed to receiver capacity, reducing congestion and improving throughput.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data.
During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

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5. Session Layer Synchronization

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints before failure: 60,
120, 180, 240, 300, 360

Total checkpoints:
6

Retransmission:

$$390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = **6**
- Data retransmitted = **30 MB**

Importance

Synchronization allows communication to resume from the last checkpoint instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

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6. Presentation Layer

Given

- Original = 900 MB

- Compression = 40%
- Encryption overhead = 5%

Compressed size:

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size:

$$540 \times 1.05 = 567 \text{ MB}$$

Reduction:

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Presentation Layer

Provides compression for efficiency and encryption for security.

7.An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

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7. FTP Transfer

Given

- Data = 3 GB = 3000 MB
- Throughput = 120 Mbps

- Request size = 3 MB Requests:

$$\frac{3000}{3} = 1000$$

Bits:

$$3000 \times 8 = 24{,}000 \text{ Mb}$$

Transmission time:

$$\frac{24000}{120} = 200 \text{ s}$$

Answer

- Requests = **1000**
- Transmission time = **200 s**

Application Layer

Provides user services such as FTP, authentication, file transfer, and error reporting.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

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8. End-to-End Delay

Processing delay:

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total delay:

$15 + 18 + 12 = 45$ ms

Answer

- Total end-to-end delay = **45 ms**

Layer Contributions

- Application: User services
- Transport: End-to-end communication
- Network: Routing
- Data Link: Framing and error detection
- Physical: Bit transmission

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

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9. Frame Overhead

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes

Useful data/frame:

$1600 - 80 = 1520$ bytes

Frames:

$$\left\lceil \frac{80{,}000{,}000}{1520} \right\rceil = 52{,}632$$

Total overhead:

$52{,}632 \times 80 = 4{,}210{,}560$ bytes

Efficiency:

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**
- Overhead = **4,210,560 bytes (≈4.21 MB)**
- Efficiency = **95%**

Effect of Overhead

Higher protocol overhead reduces bandwidth efficiency and increases transmission time.

10.A hospital transfers a patient's medical record of 240 MB through an IoTenabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

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10. Healthcare Network

Given

- Original = 240 MB

- Compression = 30%
- Segment size = 1400 bytes
- Frame overhead = 60 bytes

Compressed size:

$$240 \times 0.70 = 168 \text{ MB}$$

Compressed bytes:

$168{,}000{,}000$ bytes

Segments:

$$\frac{168{,}000{,}000}{1400} = 120{,}000$$

Protocol overhead:

$120{,}000 \times 60 = 7{,}200{,}000$ bytes $= 7.2$ MB

Final transmitted:

$$168 + 7.2 = 175.2 \text{ MB}$$

Answer

- Compressed file = **168 MB**
- Segments = **120,000**
- Protocol overhead = **7.2 MB**
- Final transmitted data = **175.2 MB**

OSI Layer Cooperation

- **Presentation Layer:** Compression and encryption.
- **Transport Layer:** Segmentation and reliable delivery.
- **Data Link Layer:** Framing and error detection.

- **Network Layer:** Routing.
- **Physical Layer:** Transmission of bits over the medium.